

Exercise 2: Verifying Interactions

ExternalApi.java:

```
package com.exercise2;

public interface ExternalApi {

    String getData();

}
```

MyService.java:

```
package com.exercise2;

public class MyService {

    private final ExternalApi externalApi;

    public MyService(ExternalApi externalApi) {

        this.externalApi = externalApi;

    }

    public String fetchData() {

        return externalApi.getData();

    }

}
```

MyServiceTest.java:

```
package com.exercise2;
```

```
import static org.mockito.Mockito.*;

import org.junit.jupiter.api.Test;

import org.mockito.Mockito;

public class MyServiceTest {

    @Test
    public void testVerifyInteraction() {
        // Step 1: Create a mock object
        ExternalApi mockApi = Mockito.mock(ExternalApi.class);
        MyService service = new MyService(mockApi);

        // Step 2: Call the method with specific arguments
        service.fetchData();

        // Step 3: Verify the interaction - verify that getData() was called
        verify(mockApi).getData();
    }

    @Test
    public void testVerifyInteractionWithTimes() {
        ExternalApi mockApi = Mockito.mock(ExternalApi.class);
        MyService service = new MyService(mockApi);

        service.fetchData();
```

```

        // Verify that getData() was called exactly once

        verify(mockApi, times(1)).getData();
    }

@Test

public void testVerifyNeverCalled() {

    ExternalApi mockApi = Mockito.mock(ExternalApi.class);

    MyService service = new MyService(mockApi);

    // Verify that getData() was never called

    verify(mockApi, never()).getData();

}

}

```

Output:

```

PS D:\SD\Digital-Nurture-JavaFSE-main\local repo\Cognizant-exercices\Week 1\Mockito\Exercise-2\exercise2> mvn clean test
[INFO] T E S T S
[INFO] -----
[INFO] Running com.exercise2.MyServiceTest
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::objectFieldOffset has been called by net.bytebuddy.dynamic.loading.ClassInjector$UsingUnsafe$Dispatcher$CreationAction (file
:/C:/Users/adity/.m2/repository/net/bytebuddy/byte-buddy/1.14.9/byte-buddy-1.14.9.jar)
WARNING: Please consider reporting this to the maintainers of class net.bytebuddy.dynamic.loading.ClassInjector$UsingUnsafe$Dispatcher$CreationAction
WARNING: sun.misc.Unsafe::objectFieldOffset will be removed in a future release
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.550 s -- in com.exercise2.MyServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.419 s
[INFO] Finished at: 2026-06-21T21:37:49+05:30
[INFO]

```